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TEXAS
The University of Texas at Austin

GENERAL LEVEL AND DAILY FLUCTUATIONS IN DISCRIMINATION INTERACT TO SHAPE DAILY CORTISOL RHYTHMS IN MEXICAN-ORIGIN YOUTH

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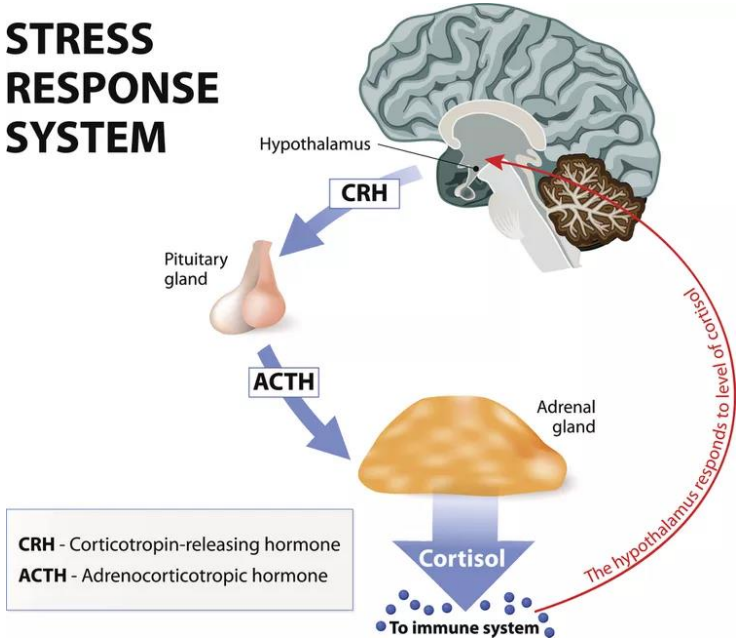
INTRODUCTION

BACKGROUNDS

- Mexican-origin youth is facing heightened discrimination in sociopolitical climates characterized by anti-immigrant sentiment (Herda, 2025).
- Such discrimination may occur on a day-to-day basis. Such as being treated with less courtesy or receiving poorer service on a specific day.
- Biopsychosocial model of minority health (Myers, 2009): daily discrimination can "get under the skin" to influence adolescents' biological health.

BACKGROUNDS

Hypothalamic-Pituitary-Adrenal (HPA) axis one of the primary biological systems involved in stress response. Cortisol is the end product (Smith & Vale, 2006).



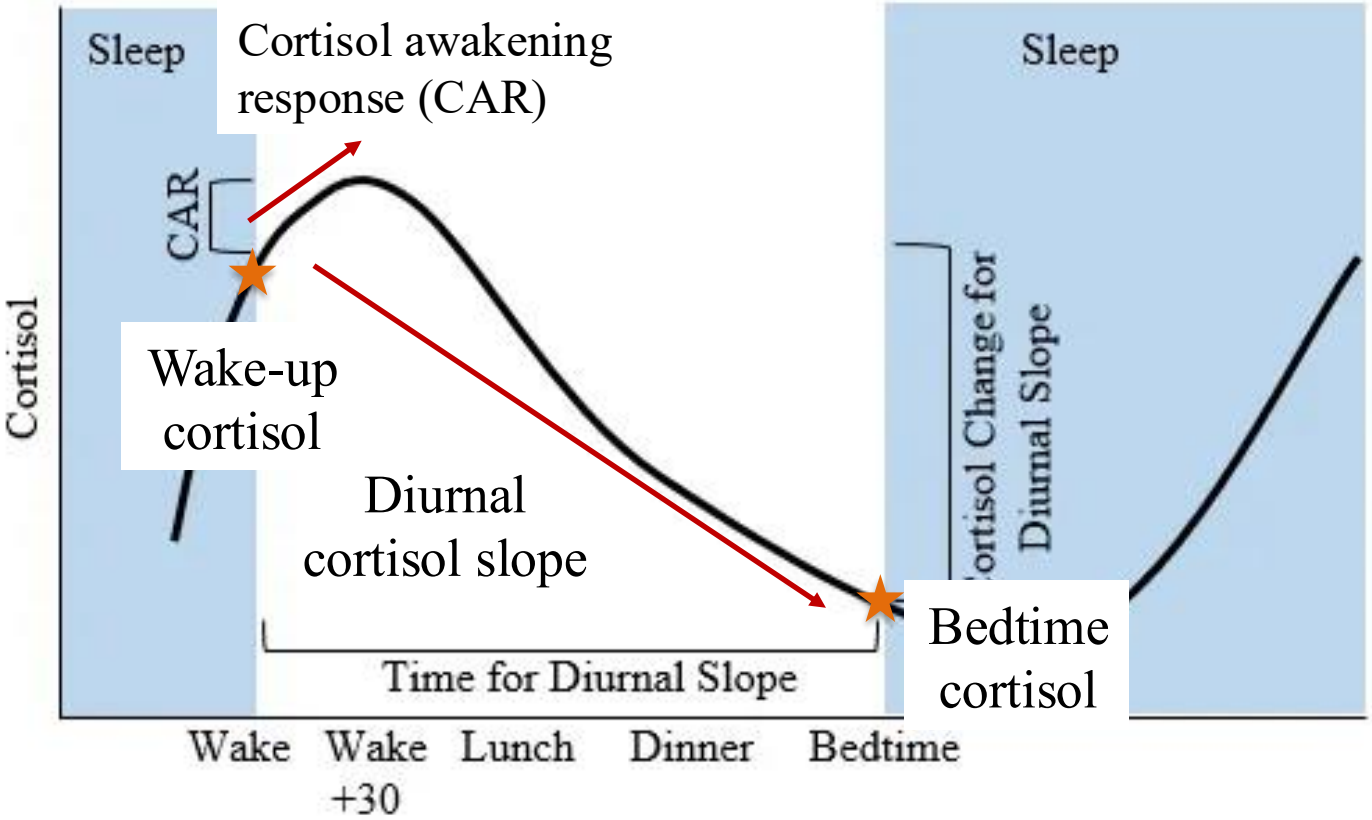
Hypothalamus: detect stress signals & releases corticotropin-releasing hormone.

Pituitary gland: start releasing adrenocorticotropic hormone.

Adrenal glands: get ACTH & start producing cortisol and other substances.

BACKGROUNDS

Normal diurnal cortisol rhythm (Adam & Kumari, 2009):



PRIOR RESEARCH

Prior studies used multilevel modeling approach to learn daily discrimination influence cortisol rhythm/parameters

- Between-person (general discrimination exposure; person A vs. person B)
- Within-person (more/less discrimination on a day; deviation from person A own average)

Multilevel modeling approach yield mixed findings:

- Between-person vs. within-person
- Within the same level
- Inconsistency exits for almost all cortisol parameters.

PRIOR RESEARCH

		WAKEUP	CAR	SLOPE	BEDTIME
Chen et al., 2024	WP level	Not significant	Not significant	Not significant	Not significant
	BP level	Not significant	Not significant	higher-steeper	higher-lower (next day)
Wang et al., 2025	WP level	Not significant	higher-steeper (next day)	higher-steeper	higher-lower
	BP level	higher-higher (next day)	Not significant	higher-steeper (same day & next day)	Not significant
Sasser et al., 2024	WP level	Not significant	Not significant	Not significant	no test
	BP level	Not significant	Not significant	Not significant	no test
Seaton & Zeiders, 2021	WP level	higher-higher (next day)	higher-flatter (next day)	higher-steeper (next day)	higher-higher
	BP level	no test	no test	no test	no test
Zeiders et al., 2014	WP level	no test	no test	Not significant	no test
	BP level	no test	no test	higher-flatter	no test
Zeiders et al., 2012	WP level	no test	no test	no test	no test
	BP level	Not significant	higher-steeper	no test	Not significant

RESEARCH GAP

- Prior studies examined the within- and between- person level effects separately

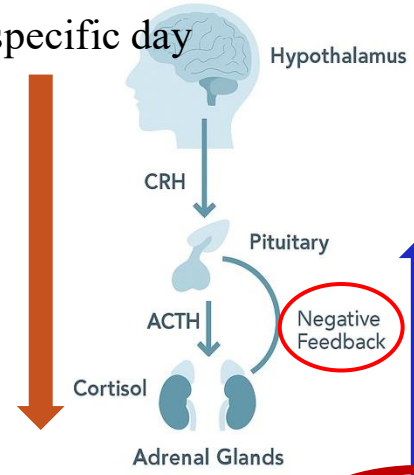
- Explore individual's cortisol response to daily discrimination fluctuation (WP level) without accounting for their general exposure (BP level).
  We studied adolescents' daily stress (discrimination) response without understanding where they came from (encounter discrimination often or not).

- Daily discrimination fluctuations and general discrimination exposure jointly shape cortisol responses (Lengton et al., 2026; Agorastos & Chrousos, 2022)

RESEARCH GAP

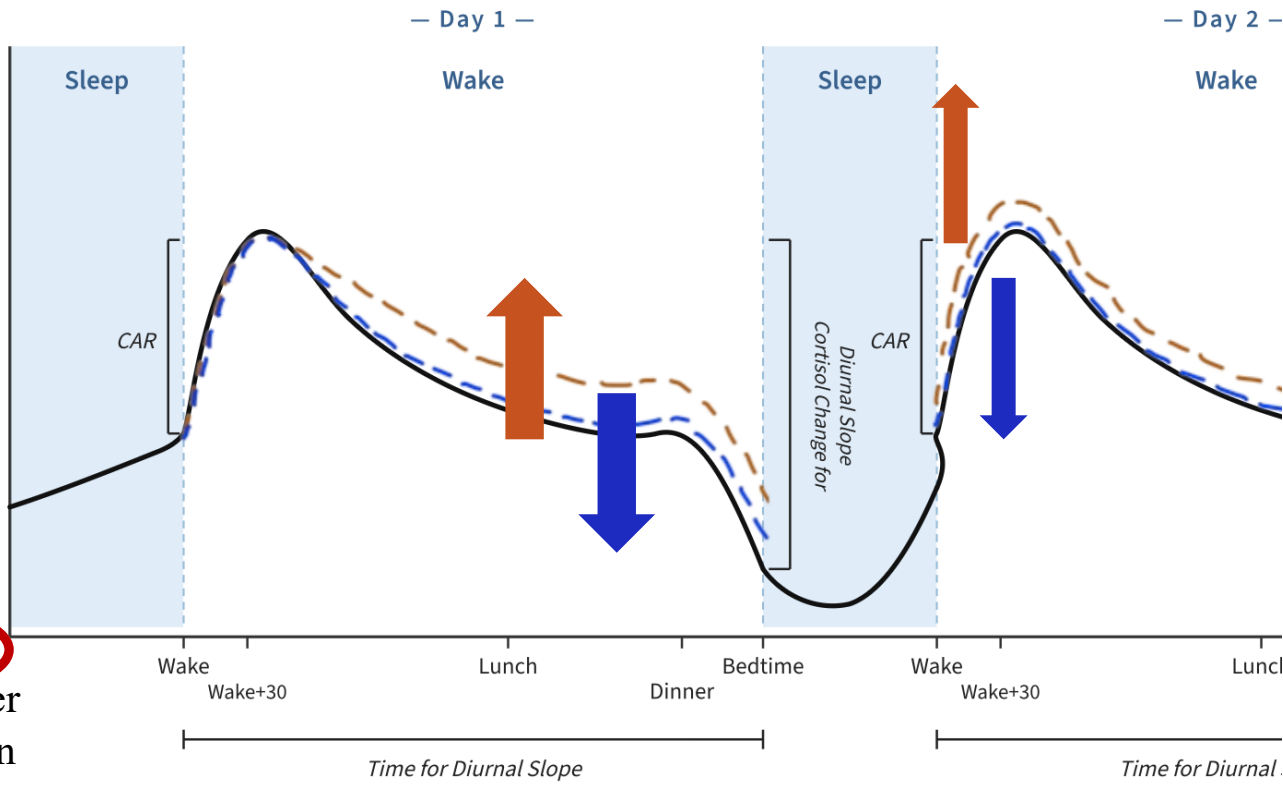
WP level

Higher discrimination on a specific day



BP level

General higher discrimination



(Agorastos & Chrousos, 2022; Kudielka & Wüst, 2010; Ostrander et al., 2006)

CURRENT STUDY

The interaction of between-person and within-person levels of discrimination on adolescents' cortisol responses

Hypothesis 1 Higher discrimination on a specific day (WP level) relates to elevated HPA-axis response

On days when adolescents experienced higher levels of discrimination compared to their own average they would exhibit higher same-day bedtime cortisol, a flatter same-day cortisol slope, higher next-day waking cortisol, and an exaggerated next-day CAR.

Hypothesis 2 Hypothesis 1 only happen among those who are generally less exposed to discrimination (BP level)

Adolescents who reported higher levels of discrimination across the study period (i.e., higher between-person level) may not show these elevated responses following daily increases in discrimination.

METHODS

Methods-Participants

- The data for this study come from a three-wave longitudinal study (Wave 1: 2012–2015, Wave 2: 2013–2016, Wave 3: 2017–2020) examining Mexican immigrant families in central Texas (W1: $N = 604$, W2: $N = 483$, W3: $N = 334$)

- At Wave 3, 290 adolescents participated in a **four-day daily diary study**:

1) A nightly survey on a tablet (daily discrimination & activities)
2) Salivary samples 3 times/day: waking, 30-45 minutes post-waking, bedtime

- 15 were excluded due to the use of steroid-based medication, **275** left
- 256 (88.5%) completed the survey on all four days
- 245 (84.8%) provided cortisol data for all four days

Methods-Measures

Daily Discrimination

Four items adapted from the Daily Discrimination Scale (Kessler et al., 1999).
Sample item: "I was treated with less courtesy than other people today."
1 (strongly disagree) to 5 (strongly agree). $\omega_{between} = 0.96$; $\omega_{within} = 0.76$

Salivary cortisol

Salivary samples were collected using SaliCaps (Passive drool)
Three times a day (winsorized at 50 nmol/L)
4 cortisol indicators:

- Bedtime cortisol Same-day Positively skewed (7.47), log transformed
- Cortisol slope: (Waking cortisol level - Bedtime cortisol level)/ Time Same-day
- Waking cortisol Next-day
- Cortisol awakening response (CAR): (30 to 45 post waking cortisol level - waking cortisol level)/ Time Next-day

Methods-Measures

Covariates

Within-person level:

- Consumption of alcohol/nicotine/caffeine (1= yes, 0= no),
- Daily exercise (1= yes, 0= no)
- Waking time
- Whether took nonsteroid-based medication in the past 12 hours (1= yes, 0= no)

Between-person level:

- Age
- Gender (0= male, 1= female)
- Nativity (0= Mexico-born, 1= US-born)
- Maternal education level: 1(Elementary and below) - 4(College/University +)

Methods-Analytic strategy

- **Person-mean centered** multilevel modeling was conducted in R using the lme4 package (Bates et al., 2003).

Main effect model

Level 1 (Within-person level): $y_{ti} = \beta_{0i} + \beta_{1i}(x_{ti} - \bar{X}_i) + (\beta_{2i,3i,4i\ldots} Daily\ covariance) + e_{ti}$

Level 2 (Between-person level):

$\beta_{0i} = \gamma_{00} + \gamma_{01}\bar{X}_i + (\gamma_{02,03,04\ldots}\ Person\text{-}level\ Covariates) + U_{0i}$ (note: random intercept)

$\beta_{1i} = \gamma_{10} + U_{1i}$ (note: random slope)

Composite:

$y_{ti} = \gamma_{00} + \gamma_{01}\bar{X}_i + \gamma_{10}(x_{ti} - \bar{X}_i) + U_{1i}(x_{ti} - \bar{X}_i) + (\beta_{3i,4i,5i\ldots} Daily\ covariance) + (\gamma_{02,03\ldots}\ Person - level\ Covariates) + U_{0i} + e_{ti}$

Interaction model

Level 1 (Within-person level): $y_{ti} = \beta_{0i} + \beta_{1i}(x_{ti} - \bar{X}_i) + (\beta_{3i,4i,5i\ldots} Daily\ covariance) + e_{ti}$

Level 2 (Between-person level):

$\beta_{0i} = \gamma_{00} + \gamma_{01}\bar{X}_i + (\gamma_{02,03,04\ldots}$ Person-level Covariates$) + U_{0i}$ (note: random intercept)

$\beta_{1i} = \gamma_{10} + \gamma_{11}\bar{X}_i + U_{1i}$ (note: random slope)

Composite:

$y_{ti} = \gamma_{00} + \gamma_{01}\bar{X}_i + \gamma_{10}(x_{ti} - \bar{X}_i) + \boldsymbol{\gamma_{11}\bar{X}_i(x_{ti} - \bar{X}_i)} + U_{1i}(x_{ti} - \bar{X}_i) + (\beta_{3i,4i,5i\ldots} Daily\ covariance) + (\gamma_{02,03,04\ldots}$ Person − level Covariates$) + U_{0i} + e_{ti}$

Methods-Analytic strategy

- Four models were fitted separately considering the multicollinearity issues (i.e., diurnal slope was derived from waking and bedtime cortisol)
- 62 cases in which sample 2 (30-post waking) fell outside the 45-minute window after sample 1, CARs were coded as missing (Stalder et al., 2022).
- For each model, listwise deletion handles missing data on both predictors and outcomes (Chen et al., 2024; Kische et al., 2021; Seaton & Zeiders, 2021)

RESULTS

Descriptive statistics

Table 1 Descriptive information for same-day effect samples

	Overall (*N*=275)
Adolescent age	
Mean (SD)	17.5 (0.938)
Median [Min, Max]	17.4 [15.0, 21.0]
Missing	5 (1.8%)
Adolescent gender	
Male	124 (45.1%)
Female	151 (54.9%)
Adolescent nativity	
U.S. born	208 (75.6%)
non-U.S. born	67 (24.4%)
Mother's education level	
Elementary and below	76 (27.6%)
Middle school	89 (32.4%)
High school	56 (20.4%)
College/University +	49(17.8)
Missing	5 (1.8%)
First_day_alcohol_consumption	
consumed	7 (2.5%)
no consumption	259 (94.2%)
Missing	9 (3.3%)
First_day_nicotine_consumption	
consumed	4 (1.5%)
no consumption	268 (97.5%)
Missing	3 (1.1%)
First_day_caffin_consumption	
consumed	116 (42.2%)
no consumption	159 (57.8%)

Intraclass correlations (ICC):

- 29.6% to 63.7% of the variance in cortisol parameters was attributable to between-person (BP) differences
- 36.3% to 70.4% was attributable to within-person (WP) variation

Figure 1 The association between within-person discrimination and Wave 3 adolescents' same-day cortisol slope, influenced by between-person discrimination

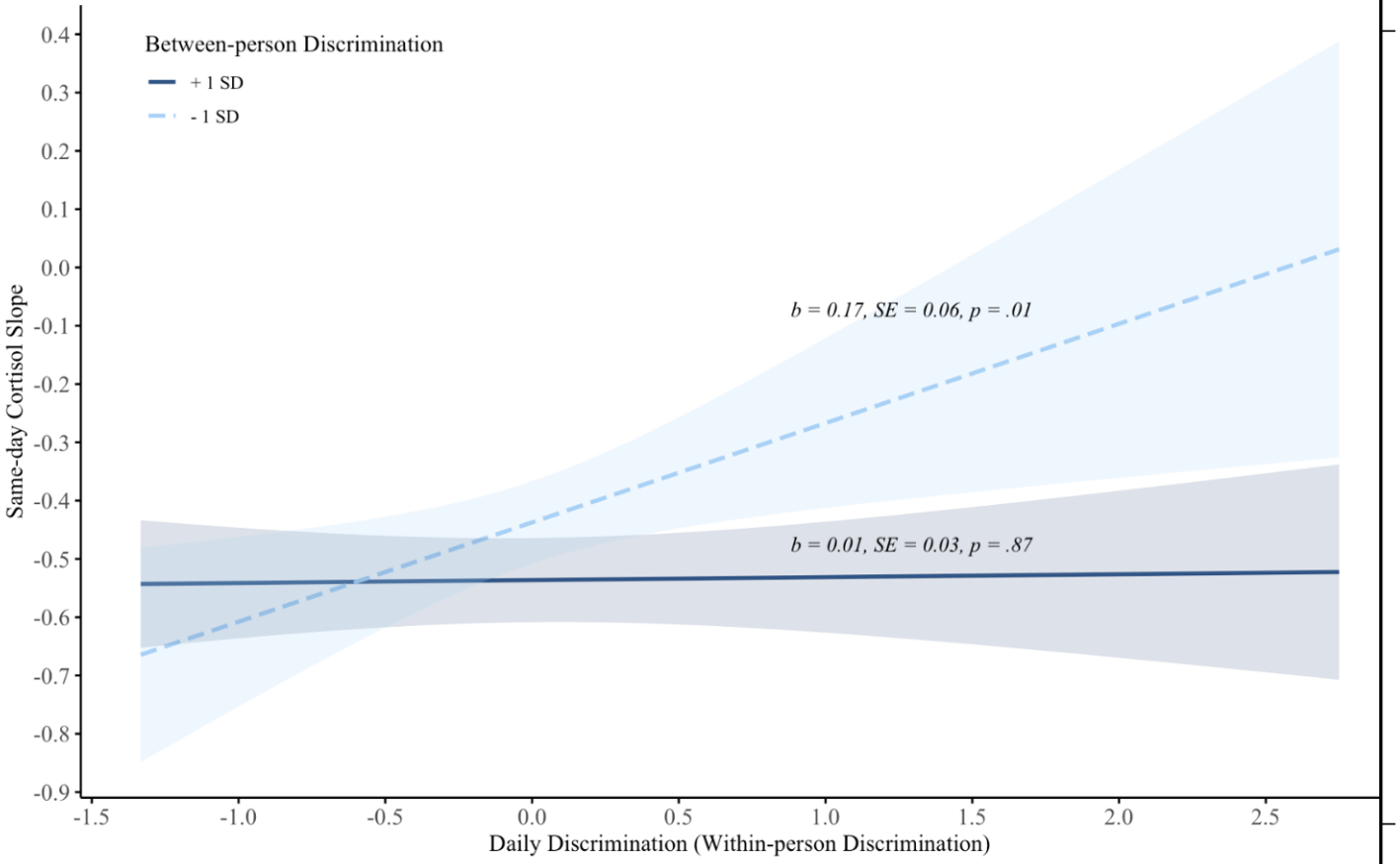
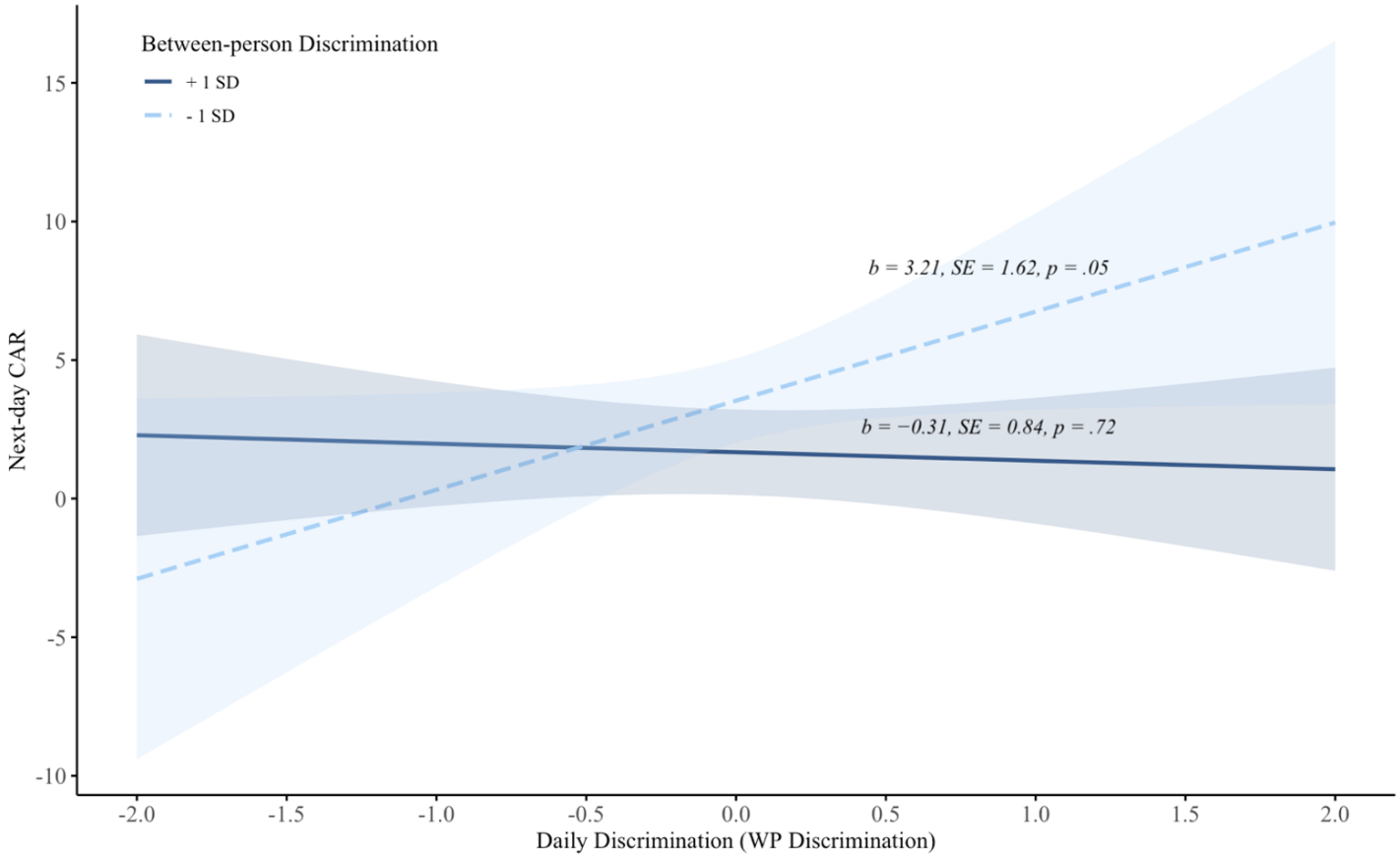


Figure 2 The association between within-person discrimination and Wave 3 adolescents' next-day CAR, influenced by between-person discrimination



KEY TAKEAWAYS

- Adolescents who are generally less exposed to discrimination show heightened same-day and next-day cortisol response (i.e., flattened same-day cortisol slope and exaggerated next-day CAR)

- Those who generally experienced more discrimination, daily increases in discrimination were not significantly associated with cortisol response.

For adolescents with lower general discrimination exposure, a day with heightened discrimination may be particularly stressful and elicit a strong HPA response.

DISCUSSION

IMPLICATION & CONTRIBUTION

- **Daily discrimination heightens, dampens, or has no effect on cortisol response coexist**
- General discrimination exposure is an important context
- The physiological consequences of daily discrimination must be situated within adolescents' broader context of exposure

- **The double-edged nature of the physiological adaptation under discrimination**
- Physiological resilience (prevent prolonged cortisol elevation) (Barthel et al., 2025)
- A sign of HPA dysregulated fatigue (cost) (Ostrander et al., 2006)

- **Interventions should pay special attention to:**

1. Adolescents who usually encounter little discrimination but occasionally face it
2. Adolescents experience discrimination chronically (blunted immediate responses in HPA axis may mask underlying stress accumulation)

LIMITATION

- Sample consisted solely of Mexican-origin adolescents from Central Texas, which may limit the generalizability of the findings.

- Focused only on the effects of discrimination on cortisol outcomes, but it also shapes psychological, academic, social adjustment ("above the skin").

- 4 days daily diary can only provide a limited snapshot of adolescents' daily lives.

↓

Adolescents' broader discrimination exposure history over years or in sensitive period (early/middle adolescents) may shape the daily cortisol response patterns

THANKS

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QUESTIONS

1. I was treated with less respect than other people today. Fui tratado/a con menos respeto que otras personas el dÃa de hoy.
2. People thought I was not smart today. La gente pensÃ³ que no era inteligente el dÃa de hoy.
3. People acted like I was dishonest today. La gente actÃºo como si yo fuera deshonesto/a el dÃa de hoy.
4. I was called names or insulted today. Me llamaron nombres negativos o me insultaron el dÃa de hoy.